

Question	Marking details	Marks available					
		AO1	AO2	AO3	Total	Maths	Prac
7	Indicative content Anaesthetic A A1 Anaesthetic A {blocks Na⁺ channels / Na⁺ channels don't open} A2 Threshold (potential) is not met/ ref to all or nothing A3 (membrane) not depolarised A4 Action potential is not generated A5 So impulse does not reach {pain centre/ brain} Anaesthetic B B1 Anaesthetic B Blocks Ca²⁺ channels/ Ca²⁺ channels do not open B2 (Ca²⁺ can't enter) so synaptic vesicles don't fuse with presynaptic membrane B3 Neurotransmitter not released into synaptic cleft/ no exocytosis B4 Postsynaptic membrane not depolarised B5 No action potential generated B6 So impulse does not reach {pain centre/ brain} Compare with reflex arc C1 Both have sensory neurones/ both have three neurones C2 (With cell bodies) in ganglia C3 Outside CNS C4 Pain pathway has no motor neurones/ shorter motor neurone/ motor neurone in brain Accept reverse answer for reflex C5 Pain pathway has no relay neurones/ relay neurone in brain/longer relay neurone Accept reverse answer for reflex C6 Pain pathway has no effector/ effector is located in the brain C7 Involves brain not spinal cord	5	4		9		

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				7-9 marks Indicative content of this level is detailed information from all three lists <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i> 4-6 marks Indicative content of this level is detailed information from two of the lists or less detail from three. <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i> 1-3 marks Indicative content of this level is some information from at least one of the lists or little information from more than one list. <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
				Question 7 total	5	4	0	9	0	0